CampusNotes - Academic Study Resource

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING - SEMESTER 5

OFFICIAL ACADEMIC STUDY MATERIAL - CSE CURRICULUM

Experiment 4: TCP Socket Programming for Client-Server Chat

Course: Advanced Java Programming Laboratory [23CS3504]
Category: Experiments / Practicals | Experiment #4 | Academic Year: 2026-27

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1. Overview & Objective:

Build a multi-client TCP network socket application supporting bi-directional text streaming.

2. Practical Tasks & Verification Steps:

1. Construct ServerSocket listening on port 8080 dispatching each incoming client connection to worker thread.
2. Build ClientSocket connecting to server with BufferedReader and PrintWriter I/O streams.
3. Implement broadcast messaging routing messages from one client to all other connected participants.
4. Handle graceful client disconnection with socket close cleanup and active client count broadcast.

Submission Guidelines & Verification:

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